

4(a)	sequence of, nucleotides / bases, that is part of DNA ; A sequence of DNA nucleotides (that) codes for a polypeptide ; A protein / enzyme / amino acid chain	2
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Question	Answer	Marks
4(b)(i)	<i>must attempt both DNA polymerase and DNA ligase to gain max</i> <i>any five from:</i> <i>DNA polymerase</i> 1 addition of, activated / phosphorylated, nucleotides ; 2 <i>ref. to</i> complementary, nucleotides / bases / strands ; 3 forms phosphodiester bonds ; 4 (between adjacent nucleotides and) elongating / growing, polynucleotide / strand / AW ; 5 <i>ref. to</i> proofreading ability of DNA polymerase ; <i>DNA ligase</i> 6 joins / AW, Okazaki fragments ; 7 on lagging strand ; 8 forms phosphodiester bonds, between the fragments / to complete phosphate-sugar backbone ;	5
4(b)(ii)	S phase / synthesis phase ; ! 1 as in S1 phase / S unqualified	1

Question	Answer	Marks
4(c)(i)	A - centromere <u>and</u> I kinetochore site of attachment of, chromatids / chromosome(s) to, spindle fibres / microtubules or holds, sister / identical, chromatids together ; R daughter chromatids B - spindle fibres / microtubules, <u>and</u> orientating chromosomes at the (spindle) equator or separating chromatids, at end of metaphase / at start of anaphase or movement of, chromatids / chromosomes, to (opposite) poles ; <i>allow 1 mark max if functions incorrect but A and B named correctly</i> <i>allow 1 mark max if functions are correct but A named as kinetochore <u>and</u> B poorly misspelt attempt at microtubules</i>	2
4(c)(ii)	<u>two</u> separate, chromatids / single-stranded chromosomes ; chromatids attached to spindle with fibres drawn to, poles / centrioles ; single chromatids drawn U-shaped or V-shaped pointing towards the poles ; centromeres drawn in both chromatids ;	3